CIVIL AND ENVIRONMENTAL ENGINEERING DEPARTMENT (CEE)

Faculty Members

Chairperson: Meheddene Machaka
Associate Professor: Meheddene Machaka, Riad Al Wardany,
Assistant Professor: Rana Hajj Chehade
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Programs Offered

Currently, the CEE Department offers one program in Civil Engineering, leading to Bachelor of Science, Bachelor of Engineering, and Master of Science degrees.

CIVIL ENGINEERING PROGRAM

Mission

The mission of the Civil & Environmental Engineering Department at Rafik Hariri University is fourfold:

- To provide high-quality education at the undergraduate and graduate levels.
- To prepare students for successful careers in the civil engineering field and research.
- To equip students with the will and tools to engage in lifelong learning, teamwork, team leading, and continual professional development.
- To cultivate in students habitual ethical and tolerant behavior and the desire to positively contribute to the advancement of their community and their profession.

Objectives

The educational objectives of the Civil Engineering program intend to enable its graduates, within a few years of graduation, to:

- Achieve successful careers in civil engineering and related fields;
- Pursue graduate studies and research in civil engineering and other fields;
- Effectively function as team members or as team leaders;
- Continue their lifelong learning process and their professional development; and
- Participate in community service and positively contribute to their society.

Program Outcomes

The Civil & Environmental Engineering Department adopts ABET Student Learning Outcomes as the Civil Engineering (CIVE) Program outcomes. Conforming to the Program Objectives, graduates of the CIVE Program will acquire each of the following abilities and characteristics:

1. Ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.
2. Ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.
3. Ability to communicate effectively with a range of audiences.
4. Ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.

5. Ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.
6. Ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.
7. Ability to acquire and apply new knowledge as needed, using appropriate learning strategies.

Accreditation

The Bachelor of Engineering program in Civil Engineering is accredited by the Engineering Accreditation Commission of ABET (https://www.abet.org) under the commission's General Criteria and Program Criteria for Civil and Similarly Named Engineering Programs.

Career Opportunities

Civil engineers attain a broad spectrum of skills sought after by a wide range of professions. Industrial sectors, enterprises, and services in which a civil engineer can pursue a career include construction, infrastructure, transportation, consulting, water resources, environmental, government, and other public works sectors. Most importantly, students will be equipped with the skills to pursue entrepreneurial ventures that enable them to realize their own ideas. There is so much work to be done to fashion a better future, and civil engineers can be major contributors to that objective.

Program Overview

The Civil Engineering Program at Rafik Hariri University puts students' best interests first and foremost. Every little experience the student gains represents a block in building a competent, confident, purposeful, problem-solving, competitive, responsible, and conscientious individual. This is accomplished through a curriculum and facilities that meet the highest standards, faculty members committed to students' academic and personal growth, and an environment that inspires learning and drives creativity.

The Bachelor of Engineering (BE) program in Civil Engineering comprises 152 credit hours across eight regular semesters and three summer semesters. The Bachelor of Science (BS) program comprises 118 credit hours across six regular semesters and three summer semesters. In addition to completing the credit requirements, a BS or a BE degree is conferred upon a student if the earned Cumulative GPA, Major GPA, and the Summative Learning Project grade are all 70 or above.

The required credit hours span three categories: University requirements (UR), College requirements (CR), and Program requirements (PR). The allocation of the credit hours for the BS and BE programs is shown in the following table:

BS Program in Civil Engineering (118 Credits)						
Courses Category	Major		Non-Major		Credits	Percent
	Mandatory	Electives	Mandatory	Electives		
General Education	0	0	20	7	27	23
College Requirement	0	0	24	0	24	21
Program Requirement	58	0	9	0	67	56
Credits	58	0	53	7	118	100

BE Program in Civil Engineering (152 Credits)						
Courses Category	Major		Non-Major		Credits	Percent
	Mandatory	Electives	Mandatory	Electives		
General Education	0	0	20	10	30	20
College Requirement	0	0	27	0	27	18
Program Requirement	71	15	9	0	95	62
Credits	71	15	56	10	152	100

I. University Requirements (General Education)

The list of the University-required courses and their descriptions is presented in the introductory pages of the College of Engineering section in this catalog.

II. College Requirements

The list of the College-required courses and their descriptions is presented in the introductory pages of the College of Engineering section in this catalog.

III. Program Requirements

The program requirements for a Bachelor of Engineering degree in Civil Engineering encompass 95 credits (67 credits for the BS degree), distributed as follows: 80 credits (67 credits for the BS degree) of mandatory courses and 15 credits of elective courses for the BE degree only. Both categories are summarized below.

A. Mandatory requirements

The mandatory courses for the major and non-major CIVE programs are listed in the table below.

Course #	Title	Credits	Prerequisites
CIVE 200	Introduction to Civil Engineering	2	
CIVE 202	Civil and Architectural Drawings	3	
CIVE 211	Statics	3	

CIVE 240	Surveying	3	
CIVE 312	Structural Analysis I	3	MECH 320
CIVE 314	Concrete I	3	MECH 320
CIVE 321	Construction Materials	3	
CIVE 321L	Construction Materials Lab	1	Co-req: CIVE 321
CIVE 341	Highway Engineering	3	CIVE 240
CIVE 342	Traffic Engineering	3	CIVE 341
CIVE 400*	Summative Learning (BS) Project	3	ENGL 217 Senior Standing
CIVE 413	Computer Modeling of Structures	3	CIVE 312
CIVE 415	Concrete II	3	CIVE 312 and CIVE 314
CIVE 417	Steel Structures	3	CIVE 312
CIVE 423	Soil Mechanics	3	MECH 320; GEOL 221
CIVE 423L	Soil Mechanics Laboratory	1	Co-req.: CIVE 423
CIVE 424	Foundation Engineering	3	CIVE 423
CIVE 432	Environmental Engineering	3	CHEM 211
CIVE 435	Hydraulics	3	MECH 333
CIVE 499	Co-op Work Experience	3	ENGL 217; Senior Standing
CIVE 501	Construction Planning and Management	3	Senior Standing
CIVE 507	Structural Analysis II	3	CIVE 312
CIVE 508	AI Applications in Civil Engineering	3	CCEE 214
CIVE 536	Hydrology	3	CIVE 435
CIVE 538	Water and Wastewater Treatment	3	CIVE 432, CIVE 435
CIVE 595A	BE Summative Learning Project 1	1	Senior Standing
CIVE 595B	BE Summative Learning Project 2	3	CIVE 595A
CIVE 5xx	Technical Electives I-V	15	Per course requirements
MECH 220	Dynamics	3	CIVE 211
MECH 320	Mechanics of Materials	3	CIVE 211
MECH 333	Thermal Fluid Sciences	3	MATH 216

* Required for BS-bound students. BE students take a technical elective instead.

The civil engineering courses cover five areas, as listed in the following table.

Code	Course Title	Credits	Prerequisite
Common Courses and Construction Management			
CIVE 200	Introduction to Civil Engineering	2	
CIVE 202	Civil and Architectural Drawings	3	
CIVE 595	BE Summative Learning Project (A + B)	3	Senior standing
CIVE 501	Construction Planning and Management	3	Senior standing
CIVE 508	AI Applications in Civil Engineering	3	CCEE 214
Structural Analysis & Design			
CIVE 211	Statics	3	
CIVE 312	Structural Analysis I	3	MECH 320
CIVE 314	Concrete I	3	MECH 320

CIVE 413	Computer Modeling of Structures	3	CIVE 312
CIVE 415	Concrete II	3	CIVE 312 and CIVE 314
CIVE 417	Steel Structures	3	CIVE 312
CIVE 507	Structural Analysis II	3	CIVE 312
Construction Materials & Geotechnics			
CIVE 321	Construction Materials	3	
CIVE 321L	Construction Materials - Laboratory	1	Co-req: CIVE 321
CIVE 423	Soil Mechanics	3	MECH 320 & GEOL 221
CIVE 423L	Soil Mechanics Laboratory	1	Co-req.: CIVE 423
CIVE 424	Foundation Engineering	3	CIVE 423
Water Resources & Wastewater Treatment			
CIVE 432	Environmental Engineering	3	CHEM 211
CIVE 435	Hydraulics	3	MECH 333
CIVE 536	Hydrology	3	CIVE 435
CIVE 538	Water and Wastewater Treatment	3	CIVE 432, CIVE 435
Public Works			
CIVE 240	Surveying	3	
CIVE 341	Highway Engineering	3	CIVE 240
CIVE 342	Traffic Engineering	3	CIVE 341

B. Technical Electives

As part of the Bachelor of Engineering in Civil Engineering program, the student is required to complete 15 credit hours of technical electives. These courses allow the student to attain in-depth knowledge and understanding in a focused area of interest. The students can mix and match elective courses from different areas to gain broader knowledge across Civil Engineering disciplines. The student should, in cooperation with the academic advisor, select the set of electives that best meets their needs and aspirations. The CIVE available technical elective courses are distributed in five areas:

1. Construction Management
2. Structural Analysis and Design
3. Construction Materials and Geotechnics
4. Water Resources & Wastewater Treatment
5. Public Works

It is highly recommended that students take elective courses from the list below after completing the mandatory requirements in their respective areas. Students may select technical electives from other programs, subject to the approval of the academic advisor and the corresponding course instructor.

Course #	Title	Credits	Prerequisites
Construction Management Sequence			
CIVE 502	Engineering Risk Analysis	3	MATH 351
CIVE 503	Computer Methods in Civil Engineering	3	Senior Standing

CIVE 504	Infrastructure Asset Management	3	Senior Standing
CIVE 506	LEED Principles in Construction Management	3	Senior Standing
Structural Analysis and Design Sequence			
CIVE 510	Finite Elements Analysis	3	CIVE 312
CIVE 511	Structural Dynamics	3	CIVE 312
CIVE 514	Earthquake Engineering	3	CIVE 312
CIVE 515	Prestressed Concrete	3	CIVE 314
CIVE 519	Tall Building Structures	3	CIVE 413
Construction Material and Geotechnics Sequence			
CIVE 521	Special Concretes	3	CIVE 321
CIVE 522	Durability of Construction Materials	3	CIVE 321
CIVE 523	Repair of Concrete Structures	3	CIVE 321
CIVE 524	Nondestructive Evaluation and Instrumentation of Infrastructure	3	CIVE 321
CIVE 525	Concrete Technology	3	CIVE 321
CIVE 526	Admixtures for Concrete	3	CIVE 321
CIVE 527	Seepage and Dams	3	CIVE 424 Co-req.: CIVE 536
CIVE 528	Earth Retaining Structures	3	CIVE 314, 424
CIVE 529	Advanced Foundation Engineering	3	CIVE 424
Water and Environmental Engineering Sequence			
CIVE 571	Water Quality	3	CIVE 432
CIVE 573	Solid Waste Management	3	CIVE 432
CIVE 575	Water Supply & Wastewater Systems	3	CIVE 435
Public Works Sequence			
CIVE 540	GIS for Civil Engineering	3	Senior Standing
CIVE 541	Pavement Design	3	CIVE 423

C. Summative Learning Project

Students must complete a 3-credit-hour course for the BS (4 credits for the BE; taken 1 credit in the first regular semester and 3 credits in the following regular semester) in which they work, preferably in groups, on a problem of concern to industry or the community at large, or to develop a promising idea. The SLP could be an extension of the projects students worked on in the Engineering for the Community course, bringing them to a more useful outcome.

D. Co-op Experience

Students must complete 8 weeks of practical training in a field related to their area of interest. The Co-op experience is usually fulfilled during the summer semester of the third year in the program. The Co-op office matches the Co-op assignment with the student's field of study and the employer's interest. A faculty member follows up on a student's progress throughout the Co-op period by conducting field visits and ensuring that the student's performance is aligned with their aspirations and the employer's needs. Students are required to submit a formal report and a poster and to give a formal

presentation about their Co-op experience. While ENGL 217 and senior standing are the essential prerequisites for co-op training, the program may stipulate additional conditions.

Study Plan

The following study plan summarizes the courses and credit distribution for the Bachelor of Science (BS) and Bachelor of Engineering (BE) in CIVE Engineering. The following study plan serves as a roadmap for smooth progression toward graduation.

Course No.	Title	Credits	Prerequisites
Year 1, Fall Semester (15 Credits)			
CCEE 214	Introduction to Programming	3	
CIVE 211	Statics	3	
ENGL 210	English Composition and Rhetoric	3	Placement
MATH 215	Calculus III	3	
PHYS 210	Fundamentals of Physics and Lab	3	
Year 1, Spring Semester (17 Credits)			
ARAB 212	Arabic Language & Communication	2	
MATH 216	Calculus IV	3	MATH 215
CIVE 200	Introduction to Civil Engineering	2	
	Community & Sustainability	1	Co-req.: ENGL 210
GEOL 221	Fundamentals of Geology	3	
MECH 320	Mechanics of Materials	3	CIVE 211
	Social Sciences	3	
Year 1, Summer Semester (9 Credits)			
CHEM 211	Environmental Chemistry and Lab	3	
CIVE 240	Surveying	3	
MATH 311	Linear Algebra and Applications	3	
Year 2, Fall Semester (16 Credits)			
CIVE 312	Structural Analysis I	3	MECH 320
CIVE 321	Construction Materials	3	
CIVE 321L	Construction Materials - Laboratory	1	Co-req: CIVE 321
CIVE 341	Highway Engineering	3	CIVE 240
MATH 314	Ordinary Differential Equations	3	MATH 216
MECH 220	Dynamics	3	CIVE 211
Year 2, Spring Semester (18 Credits)			
CIVE 202	Civil and Architectural Drawings	3	
CIVE 314	Concrete I	3	MECH 320
CIVE 342	Traffic Engineering	3	CIVE 341
ENGL 217	Professional English	3	ENGL 210
MATH 351	Probability and Statistics	3	MATH 215
MECH 333	Thermal Fluid Sciences	3	MATH 216

Year 2, Summer Semester (9 Credits)			
ELEC 210	Electric Circuits	3	PHYS 191 or Equivalent
ENGR 300	Engineering Economics & Management	3	
	Humanities / Fine Arts Elective	3	
Year 3, Fall Semester (16 Credits)			
CIVE 413	Computer Modeling of Structures	3	CIVE 312
CIVE 415	Concrete II	3	CIVE 312 and CIVE 314
CIVE 423	Soil Mechanics	3	MECH 320 & GEOL 221
CIVE 423L	Soil Mechanics Laboratory	1	Co-req.: CIVE 423
CIVE 435	Hydraulics	3	MECH 333
CIVE 432	Environmental Engineering	3	CHEM 211
Year 3, Spring Semester (15 Credits)			
CIVE 417	Steel Structures	3	CIVE 312
CIVE 424	Foundation Engineering	3	CIVE 423
CIVE 507	Structural Analysis II	3	CIVE 312
MATH 421	Numerical Analysis	3	MATH 311
	Engineering Technical Elective I*	3	Per course requirements
Year 3, Summer Semester (3 Credits)			
CIVE 499	Co-op Work Experience	3	ENGL 217; Senior Standing
The above 118 credits complete the requirements for a Bachelor of Science degree in Civil Engineering.			
Year 4, Fall Semester (16 Credits)			
CIVE 501	Construction Planning and Management	3	Senior standing
CIVE 595A	BE Summative Learning Project 1	1	ENGL 217, senior standing
ENGR 510	Engineering Project Management	3	ENGR 300
	Engineering Technical Elective II	3	Per course requirements
	Engineering Technical Elective III	3	Per course requirements
CIVE 508	AI Applications in Civil Engineering	3	CCEE 214
Year 4, Spring Semester (18 Credits)			
CIVE 536	Hydrology	3	CIVE 435
CIVE 538	Water and Wastewater Treatment	3	CIVE 432, CIVE 435
CIVE 595B	BE Summative Learning Project 2	3	CIVE 595A
	Engineering Technical Elective IV	3	Per course requirements
	Engineering Technical Elective V	3	Per course requirements